

Organic Chemistry Practical: Qualitative Analysis & Functional Group Identification

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Carbonyl Identification using 2,4-Dinitrophenylhydrazine (Brady's Reagent)

Cambridge Syllabus Reference: 9701/31/M/J/24/Q3 · [10 Marks]

Three drops of an unknown liquid are added to 1 cm³ of 2,4-DNPH solution in a test-tube. A bright orange-yellow precipitate forms immediately.

(a) Identify the functional group class confirmed by this observation. [2]

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(b) Identify the type of reaction between 2,4-DNPH and a carbonyl compound. [2]

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(c) Describe how the purified precipitate can be used to identify the exact carbonyl compound. [4]

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(d) State one safety precaution when handling 2,4-dinitrophenylhydrazine. [2]

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Question 2: Discrimination Between Aldehydes and Ketones using Tollens' Reagent**Cambridge Syllabus Reference: 9701/32/M/J/23/Q3 · [10 Marks]**

Tollens' reagent is prepared by adding dilute NaOH to aqueous AgNO₃ until a brown precipitate forms, then adding dilute NH₃ dropwise until the precipitate just redissolves. Organic samples are added and heated in a 60°C water bath.

(a) State the observations when Tollens' reagent is warmed with: (i) an aldehyde, (ii) a ketone. [3]

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(b) State the formula of the active silver species in Tollens' reagent. [2]

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(c) Write the half-equation for the reduction of the silver species. [2]

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(d) Explain why Tollens' reagent residues must be washed away immediately after the practical session. [3]

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Question 3: Fehling's Alkaline Copper(II) Test for Aliphatic Aldehydes**Cambridge Syllabus Reference: 9701/34/O/N/22/Q3 · [10 Marks]**

Fehling's solution is a mixture of aqueous copper(II) sulfate (Fehling's A) and alkaline potassium sodium tartrate (Fehling's B). When warmed in a boiling water bath with ethanal, the royal blue solution turns orange-brown.

(a) State the formula and color of the precipitate formed in a positive Fehling's test. [2]

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(b) State the oxidation state change of copper in this reaction. [2]

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(c) Identify the organic oxidation product formed from ethanal in this alkaline medium. [2]

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(d) State how the reaction with Fehling's solution distinguishes aliphatic aldehydes from aromatic aldehydes (such as benzaldehyde). [4]

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Question 4: Classification of Alcohols using Acidified Potassium Dichromate(VI)**Cambridge Syllabus Reference: 9701/35/M/J/23/Q3 · [10 Marks]**

Three isomeric alcohols, A, B, and C (formula $C_4H_{10}O$), are warmed with acidified $K_2Cr_2O_7$ at $60^\circ C$.

- Alcohol A turns green, and its distillate oxidizes Tollens' reagent.
- Alcohol B turns green, but its distillate has no reaction with Tollens'.
- Alcohol C remains orange.

(a) Classify alcohols A, B, and C as primary, secondary, or tertiary. [3]

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(b) Identify the species responsible for the green color and state its oxidation number. [2]

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(c) Explain why tertiary alcohols resist oxidation by acidified dichromate. [3]

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(d) Deduce the structural formula and systematic name of alcohol C. [2]

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Question 5: Sodium Metal Effervescence Test for Alcohols and Diagnostic Hydrogen Gas Test

Cambridge Syllabus Reference: 9701/31/O/N/23/Q3 · [10 Marks]

A small freshly cut pellet of dry sodium metal is dropped into 2 cm³ of dry ethanol in a test-tube. Vigorous effervescence is observed and the tube warms up.

(a) Describe the laboratory test to confirm the identity of the gas evolved. [2]

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(b) Write the balanced chemical equation for the reaction of sodium with ethanol. [3]

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(c) Name the ionic organic salt formed in this reaction. [2]

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(d) Explain why the ethanol sample and glassware must be completely dry before performing this test. [3]

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Question 6: The Tri-iodomethane (Iodoform) Reaction for Methyl Carbonyl and Methyl Carbinol Groups

Cambridge Syllabus Reference: 9701/33/M/J/23/Q3 · [10 Marks]

Organic liquids are warmed with aqueous iodine in sodium hydroxide solution (alkaline I₂). A positive result yields a pale yellow crystalline precipitate with a distinctive antiseptic medicinal odor.

(a) State the formula and chemical name of the yellow precipitate. [2]

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(b) Identify the two specific structural features that give a positive tri-iodomethane reaction. [3]

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(c) State which of the following gives a positive test: (i) propan-1-ol, (ii) propan-2-ol, (iii) pentan-3-one, (iv) butanone. [3]

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(d) Write the balanced chemical equation for the reaction of propan-2-ol with iodine in aqueous sodium hydroxide. [2]

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Question 7: Electrophilic Addition and Unsaturation Testing with Aqueous Bromine**Cambridge Syllabus Reference: 9701/34/M/J/24/Q3 · [10 Marks]**

Bromine water, $\text{Br}_2(\text{aq})$ (orange-brown), is shaken with an unknown hydrocarbon, Liquid X. The orange color decolourizes instantly to form a colorless mixture without any catalyst.

(a) Deduce the functional group present in Liquid X. [2]

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(b) Identify the type and mechanism of this reaction. [2]

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(c) If Liquid X is cyclohexene, write the structural formula of the major product formed when reacted with aqueous bromine. [3]

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(d) Explain why alkanes (such as hexane) do not decolourize bromine water in the dark at room temperature. [3]

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Question 8: Diagnostic Differentiation of Carboxylic Acids from Phenols and Alcohols**Cambridge Syllabus Reference: 9701/31/M/J/22/Q3 · [10 Marks]**

Three organic compounds containing the -OH group are tested: ethanol, phenol, and propanoic acid. Solid sodium hydrogencarbonate, NaHCO₃, is added to separate samples.

(a) State which of the three compounds produces effervescence with solid NaHCO₃. [2]

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(b) Explain in terms of acid strength (pK_a) why propanoic acid reacts with NaHCO₃ while phenol and ethanol do not. [4]

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(c) Describe how the evolved gas is confirmed as carbon dioxide. [2]

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(d) Write the ionic equation for the reaction of propanoic acid with hydrogencarbonate ions. [2]

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Question 9: Kinetics of Nucleophilic Hydrolysis of 1-Halogenoalkanes and Bond Enthalpy Ranking

Cambridge Syllabus Reference: 9701/32/M/J/24/Q3 · [10 Marks]

Equal volumes of 1-chlorobutane, 1-bromobutane, and 1-iodobutane in ethanol are placed in test-tubes in a 50°C water bath. Aqueous silver nitrate is added simultaneously to all three tubes and the time taken for precipitation is recorded.

- (a) Rank the three halogenoalkanes in order of increasing time taken for precipitate to appear (fastest to slowest). [3]

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- (b) Explain this rate sequence by referring to carbon-halogen bond enthalpies from the Data Booklet. [4]

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- (c) Explain the role of ethanol in this reaction mixture. [2]

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- (d) Write the ionic equation for the hydrolysis of 1-bromobutane by water producing bromide ions. [1]

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Question 10: Microscale Synthesis and Olfactory Identification of Esters**Cambridge Syllabus Reference: 9701/35/O/N/23/Q3 · [10 Marks]**

10 drops of glacial ethanoic acid and 10 drops of ethanol are mixed with 3 drops of concentrated sulfuric acid. The mixture is warmed in a hot water bath at 60°C for 5 minutes, then poured into a beaker containing aqueous sodium carbonate.

(a) State the role of concentrated sulfuric acid in this reaction. [2]

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(b) Explain why the reaction mixture is poured into aqueous sodium carbonate. [3]

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(c) State the characteristic olfactory observation that confirms ester formation. [2]

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(d) Write the balanced chemical equation for the formation of ethyl ethanoate, showing structural formulas. [3]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Organic Chemistry Practical: Qualitative Analysis & Functional Group Identification

Question 1: Carbonyl Identification using 2,4-Dinitrophenylhydrazine (Brady's Reagent) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Carbonyl group / aldehyde or ketone ($\text{C}=\text{O}$) [2].	[2]
(b)	Condensation / addition-elimination reaction [2].	[2]
(c)	Filter and wash the precipitate with cold ethanol [1]; recrystallize from minimum volume of hot solvent [1]; dry and determine the sharp melting point using a melting point apparatus [1]; compare the experimental melting point with literature melting point tables of 2,4-dinitrophenylhydrazone derivatives [1].	[4]
(d)	2,4-DNPH is harmful, a skin sensitizer, and stains skin and clothing yellow; wear nitrile gloves and handle in a well-ventilated area [2].	[2]

Question 2: Discrimination Between Aldehydes and Ketones using Tollens' Reagent [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	(i) Aldehyde: A shiny silver mirror coats the inner glass wall of the tube (or dark grey precipitate of silver metal) [2]; (ii) Ketone: No reaction / solution remains clear and colorless [1].	[3]
(b)	$[\text{Ag}(\text{NH}_3)_2]^+$ (diamminesilver(I) complex ion) [2].	[2]
(c)	$[\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^- \rightarrow \text{Ag}(\text{s}) + 2\text{NH}_3$ [2] (or $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$).	[2]
(d)	On standing or drying out, Tollens' solutions form silver fulminate (or silver nitride), which is dangerously shock-sensitive and explosive [3].	[3]

Question 3: Fehling's Alkaline Copper(II) Test for Aliphatic Aldehydes [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Copper(I) oxide, Cu_2O [1]; brick-red / orange-brown precipitate [1].	[2]
(b)	Copper is reduced from +2 (in blue Cu^{2+} tartrate complex) to +1 (in Cu_2O) [2].	[2]
(c)	Ethanoate ion, CH_3COO^- (or sodium ethanoate) [2].	[2]
(d)	Aliphatic aldehydes (e.g. ethanal, propanal) readily reduce Fehling's solution to red Cu_2O [2]; aromatic aldehydes (benzaldehyde) are weaker reducing agents and fail to reduce Fehling's solution (remains blue) [2].	[4]

Question 4: Classification of Alcohols using Acidified Potassium Dichromate(VI) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	A is primary (1°) [1]; B is secondary (2°) [1]; C is tertiary (3°) [1].	[3]
(b)	Chromium(III) ion, $\text{Cr}^{3+}(\text{aq})$ [1]; oxidation number = +3 [1].	[2]
(c)	The carbon atom bonded to the -OH group is attached to three other carbon atoms and carries no hydrogen atom (no alpha-hydrogen) [1.5]; oxidation would require breaking strong C-C bonds, which does not occur under mild conditions [1.5].	[3]
(d)	$(\text{CH}_3)_3\text{COH}$ [1]; 2-methylpropan-2-ol [1].	[2]

Question 5: Sodium Metal Effervescence Test for Alcohols and Diagnostic Hydrogen Gas Test [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Insert a lighted wooden splint into the neck of the tube: burns with a characteristic squeaky 'pop' (hydrogen gas, H_2) [2].	[2]

(b)	$2\text{CH}_3\text{CH}_2\text{OH} + 2\text{Na} \rightarrow 2\text{CH}_3\text{CH}_2\text{ONa} + \text{H}_2$ [3] (1 mark for correct reactants, 1 mark for sodium ethoxide, 1 mark for balancing).	[3]
(c)	Sodium ethoxide [2].	[2]
(d)	Sodium reacts violently with water: $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ [1.5]; any trace water present will give a false positive effervescence and can cause dangerous spitting or fire [1.5].	[3]

Question 6: The Tri-iodomethane (Iodoform) Reaction for Methyl Carbonyl and Methyl Carbinol Groups [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	CHI_3 [1]; tri-iodomethane (iodoform) [1].	[2]
(b)	1. $\text{CH}_3\text{-C=O}$ (methyl carbonyl / ethanoyl group attached to H or carbon) [1.5]; 2. $\text{CH}_3\text{-CH(OH)-}$ (methyl carbinol group attached to H or carbon) [1.5].	[3]
(c)	Propan-2-ol [1] and butanone [1] give positive tests (contain $\text{CH}_3\text{-CH(OH)-}$ and $\text{CH}_3\text{-C=O}$ respectively); propan-1-ol and pentan-3-one give negative tests [1].	[3]
(d)	$\text{CH}_3\text{CH(OH)CH}_3 + 4\text{I}_2 + 6\text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{CHI}_3 + 5\text{NaI} + 5\text{H}_2\text{O}$ [2].	[2]

Question 7: Electrophilic Addition and Unsaturation Testing with Aqueous Bromine [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Carbon-carbon double bond / alkene (C=C) / unsaturation [2].	[2]
(b)	Electrophilic addition [2].	[2]
(c)	2-bromocyclohexan-1-ol (or 1,2-dibromocyclohexane) [3] (credit either halohydrin in water or dibromoalkane).	[3]
(d)	Alkanes contain only strong, non-polar single sigma C-C and C-H bonds with no electron-rich pi bonds to polarize halogen molecules [1.5]; free-radical substitution requires ultraviolet (UV) light energy to homolytically cleave Br-Br bonds [1.5].	[3]

Question 8: Diagnostic Differentiation of Carboxylic Acids from Phenols and Alcohols [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Propanoic acid only [2].	[2]
(b)	Propanoic acid is a moderately weak acid ($\text{pK}_a \sim 4.9$), stronger than carbonic acid (H_2CO_3 , $\text{pK}_a \sim 6.4$) [1.5]; it donates a proton to HCO_3^- to release CO_2 gas [1]; phenol ($\text{pK}_a \sim 10$) and ethanol ($\text{pK}_a \sim 16$) are significantly weaker acids than carbonic acid and cannot protonate HCO_3^- [1.5].	[4]
(c)	Bubble gas through limewater: turns milky / forms a white precipitate [2].	[2]
(d)	$\text{CH}_3\text{CH}_2\text{COOH(aq)} + \text{HCO}_3^-(\text{s or aq}) \rightarrow \text{CH}_3\text{CH}_2\text{COO}^-(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O(l)}$ [2].	[2]

Question 9: Kinetics of Nucleophilic Hydrolysis of 1-Halogenoalkanes and Bond Enthalpy Ranking [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	1-iodobutane (fastest / yellow) < 1-bromobutane (medium / cream) < 1-chlorobutane (slowest / white) [3].	[3]
(b)	Bond enthalpies: C-Cl (338 kJ mol^{-1}) > C-Br (276 kJ mol^{-1}) > C-I (238 kJ mol^{-1}) [2]; the rate-determining step involves breaking the C-X bond [1]; C-I has the lowest bond enthalpy and requires the least activation energy to break, hydrolyzing fastest despite having the lowest polarity [1].	[4]
(c)	Acts as a mutual cosolvent: halogenoalkanes are insoluble in water, and ethanol dissolves both the organic halogenoalkane and aqueous silver nitrate to create a homogeneous mixture [2].	[2]
(d)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{H}^+ + \text{Br}^-$ [1].	[1]

Question 10: Microscale Synthesis and Olfactory Identification of Esters [10 Marks]

Part	Marking Points & Expected Observations	Marks
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(a)	Acts as an acid catalyst [1] and dehydrating agent (absorbs water to shift equilibrium towards products) [1].	[2]
(b)	To neutralize and remove excess unreacted ethanoic acid and sulfuric acid catalyst [2]; this eliminates the pungent vinegar/acid odor, making the sweet fruity smell of the ester distinct [1].	[3]
(c)	Sweet, pleasant, fruity odor / pear drops / nail varnish remover aroma [2].	[2]
(d)	$\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \rightleftharpoons \text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O}$ [3] (reversible arrow required).	[3]